AQ5.2: Activity Questions 2 - Not Graded

This assignment will not be graded and is only for practice.
Level 1

1 point
Rank of a matrix A is equal to

- [ ] the row rank of the matrix A.
- [ ] the column rank of the matrix A.
- [ ] the nullity of the matrix A.
- [ ] the number of dependent variables in the system of equations $Ax = 0$.
- [ ] the number of independent variables in the system of equations $Ax = 0$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
the row rank of the matrix A.
the column rank of the matrix A.
the number of dependent variables in the system of equations $Ax = 0$.

Let nullity of the matrix $A_{3\times 5}$ be 2. Find the rank of the matrix $A_{3\times 5}$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 3

1 point
Find the rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 2 & 4 & 6 \end{bmatrix}$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2

1 point
1 point
If A is an $m \times n$ matrix, then which of the following statements is true?

- ( ) $rank(A) + nullity(A) = m$
- ( ) $rank(A) + nullity(A) = n$
- ( ) $rank(A) + nullity(A) = mn$
- ( )

$rank(A) + nullity(A) = max\{m, n\}$

○ $rank(A) + nullity(A) = min\{m, n\}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$rank(A) + nullity(A) = n$

1 point

Which of the following is true for a homogeneous system of linear equations $Ax = 0$?

☐ It may have no solution.
☐ It can have infinitely many solutions.
☐ It can have a unique solution.
☐ Whenever it has a non-trivial(non zero) solution, it must have infinitely many solutions

No, the answer is incorrect.
Score: 0
Accepted Answers:
It can have infinitely many solutions.
It can have a unique solution.
Whenever it has a non-trivial(non zero) solution, it must have infinitely many solutions

1 point

Which of the following options are correct for a square matrix A of order $n \times n$, where n is any natural number?

☐ If the determinant is non-zero, then the nullity of A must be 0.
☐ If the determinant is non-zero, then the nullity of A may be non-zero.
☐ If the nullity of A is non-zero, then the determinant of A must be 0.
☐ If the nullity of A is non-zero, then the determinant of A may be non-zero.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If the determinant is non-zero, then the nullity of A must be 0.
If the nullity of A is non-zero, then the determinant of A must be 0.

1 point

Choose the set of correct statements.

☐ If nullity of a 3×3 matrix is c for some natural number c, $0 \leq c \leq 3$, then the nullity of $-A$ will also be c.
☐ $nullity(A + B) = nullity(A) + nullity(B)$
☐ Nullity of the zero matrix of order $n \times n$, is n.
☐ Nullity of the zero matrix of order $n \times n$, is 0.
☐ There exist square matrices A and B of order $n \times n$, such that nullity of both A and B is 0, but the nullity of $A + B$ is n.

No, the answer is incorrect.
Score: 0
Accepted Answers:

If nullity of a 3×3 matrix is c for some natural number c, $0 \leq c \leq 3$, then the nullity of $-A$ will also be c.
Nullity of the zero matrix of order $n \times n$, is n.
There exist square matrices A and B of order $n \times n$, such that nullity of both A and B is 0, but the nullity of $A + B$ is n.
The rank of an $n \times n$ matrix all of whose entries are equal to 1 is

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point

1 point
If A is a 3×4 matrix, then which of the following options are true?
- [ ] rank(A) must be less than or equal to 3.
- [ ] nullity(A) must be greater than or equal to 1.
- [ ] If A has 2 columns which are non-zero and not multiples of each other, while the remaining columns are linear combinations of these 2 columns, then $nullity(A) = 2$.
- [ ] If A has 2 columns which are non-zero and not multiples of each other, while the remaining columns are linear combinations of these 2 columns, then $nullity(A) = 1$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
rank(A) must be less than or equal to 3.
nullity(A) must be greater than or equal to 1.
If A has 2 columns which are non-zero and not multiples of each other, while the remaining columns are linear combinations of these 2 columns, then $nullity(A) = 2$.

Level 2

1 point
Let $Ax = 0$ be a homogeneous system of linear equations which has infinitely many solutions, where A is an $m \times n$ matrix (where, $m > 1, n > 1$). Which of the following statements are possible?
- [ ] $rank(A) = m$ and $m < n$.
- [ ] $rank(A) = m$ and $m > n$.
- [ ] $rank(A) = m$ and $m = n$.
- [ ] $nullity(A) = n$.
- [ ] $nullity(A) \neq 0$.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$rank(A) = m$ and $m < n$.

$nullity(A) = n$.
$nullity(A) \neq 0$.

Consider the coefficient matrix A of the following system of linear equations to answer questions 11 and 12:

$$3x_1 + 2x_2 + x_3 = 0$$

$$x_1 + x_3 = 0$$

1 point
Which one of the following vector spaces represents the null space of A appropriately?

- $\{(-t, t, t) \mid t \in R\}$
- $\{(t_1, t_2, \frac{t_2 - t_1}{2}) \mid t_1, t_2 \in R\}$
- $\{(t, -t, t) \mid t \in R\}$
- $\{(t_1, t_2, \frac{t_1 + t_2}{2}) \mid t_1, t_2 \in R\}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{(-t, t, t) \mid t \in R\}$

1 point
What will be the rank of A and nullity of A?

- $rank(A) = 3, nullity(A) = 2$
- $rank(A) = 2, nullity(A) = 1$
- $rank(A) = 1, nullity(A) = 2$
- $rank(A) = 2, nullity(A) = 0$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$rank(A) = 2, nullity(A) = 1$

Let $Ax = 0$ be a homogeneous system of linear equations which has a unique solution, where $A \in R^{n \times n}$. What is the nullity of A?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0

1 point
Suppose $x_1 \neq 0$ solves $Ax = 0$, where $A \in R^{2 \times 4}$. What is the minimum number of elements in a linearly independent subset of null space of A that also spans the set of solutions of $Ax = 0$?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point
1 point
Consider the following sets:
$S_1 = \{(1,2,3),(4,5,6),(8,9,10)\}$S1 = {(1, 2, 3), (4, 5, 6), (8, 9, 10)}
$S_2 = \{(1,2,1),(2,4,2),(1,2,5)\}$S2 = {(1, 2, 1), (2, 4, 2), (1, 2, 5)}
$S_3 = \{(1,2,3),(4,5,6),(7,8,10)\}$S3 = {(1, 2, 3), (4, 5, 6), (7, 8, 10)}
$S_4 = \{(1,2),(4,5)\}$S4 = {(1, 2), (4, 5)}
$S_5 = \{(1,2),(4,5),(3,8)\}$S5 = {(1, 2), (4, 5), (3, 8)}
Choose the set of correct options

- [ ] S_1 S1 is a basis of R^3 R3.
- [ ] S_2 S2 is a basis of R^3 R3.
- [ ] S_3 S3 is a basis of R^3 R3.
- [ ] S_4 S4 is a basis of R^2 R2.
- [ ] S_5 S5 is a basis of R^2 R2.

No, the answer is incorrect.
Score: 0
Accepted Answers:
S_3 S3 is a basis of R^3 R3.
S_4 S4 is a basis of R^2 R2.

Check Answers